

SCM Lab Report

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1. Introduction

What is Git?

- Git** is a distributed version control system used to track changes in code or files over time.
It allows multiple developers to collaborate, manage versions, and revert to previous states if needed

What is Version Control?

- Version control is a system that records changes to files over time.
It helps you track edits, updates, and different versions of a project.
You can revert back to previous versions if mistakes happen.
Tools like Git make collaboration and management easier

What is GitHub?

- GitHub** is a cloud-based platform used to store and manage code repositories online.
It uses Git to track changes and versions of projects.
It allows developers to collaborate, share, and contribute to projects easily.
Owned by Microsoft, it is widely used in software development

Difference Between Git and GitHub

Feature	Git	GitHub
Definition	Version control system	Cloud platform for hosting repositories
Type	Software tool	Web-based service
Work Location	Local machine	Online (remote server)
Internet Needed	No	Yes
Main Use	Track file/code changes	Store, share, and collaborate on projects
Collaboration	Manual (via	Built-in collaboration (pull requests, issues,

	files/servers)	etc.)
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2. Basic Git Commands

Git Configuration Commands

- Command: `git --version`

Description: Checks the installed version of Git.

- Command: `git config --global user.name "vijay"`

Description: Sets the username for Git commits globally.

- Command: `git config --global user.email "vijay.pedddinti434@gmail.com"`

Description: Sets the email for Git commits.

- Command: `git config --list`

Description: Displays all Git configuration settings.

Screenshot:

```
MINGW64:/c/Users/vijay

vijay@jarvis MINGW64 ~
$ git --version
git version 2.52.0.windows.1

vijay@jarvis MINGW64 ~
$ git config --global user.name "vijay"

vijay@jarvis MINGW64 ~
$ git config --global user.email "vijay.pedddinti434@gmail.com"

vijay@jarvis MINGW64 ~
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
init.defaultbranch=master
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
user.name="vijay"
user.email=vijay.pedddinti434@gmail.com
http.postbuffer=524288000

vijay@jarvis MINGW64 ~
$ |
```

Navigation Commands

- Command: pwd

Description: Shows current working directory.

Screenshot:

```
vijay@jarvis MINGW64 ~
$ touch nothing.cpp

vijay@jarvis MINGW64 ~
$ touch enjoy.c fun.txt

vijay@jarvis MINGW64 ~
$ pwd
/c/Users/vijay

vijay@jarvis MINGW64 ~
$ ls
AppData/
'Application Data'@
Contacts/
Cookies@
CrossDevice/
Documents/
Downloads/
Favorites/
Links/
'Local Settings'@
Music/
'My Documents'@
NTUSER.DAT
NTUSER.DAT{6e6239c7-0aea-11f1-85b4-c0a8109e56c3}.TM.blf
NTUSER.DAT{6e6239c7-0aea-11f1-85b4-c0a8109e56c3}.TMContainer000000000000000000
1.regtrans-ms
NTUSER.DAT{6e6239c7-0aea-11f1-85b4-c0a8109e56c3}.TMContainer000000000000000000
2.regtrans-ms
Nethood@
OneDrive/
PrintHood@
Recent@
'Saved Games'/
Searches/
SendTo@
'Start Menu'@
Templates@
Videos/
enjoy.c
fun.txt
nothing.cpp
ntuser.dat.LOG1
```

Command: cd folder_name

Description: Changes directory to specified folder.

Screenshot:

```

MINGW64:/c/Users/vijay/documents
Templates@
videos/
fun.txt
nothing.cpp
ntuser.dat.LOG1
ntuser.dat.LOG2
ntuser.ini

vijay@jarvis MINGW64 ~
$ cd downloads

vijay@jarvis MINGW64 ~/downloads
$ ls
ChromeSetup.exe*
Git-2.52.0-64-bit.exe*
GitHubDesktopSetup-x64.exe*
'Max Verstatppen pitstop_ f1 poster (1).jpg'
'Max Verstatppen pitstop_ f1 poster.jpg'
'Microsoft 365 Copilot Installer (1).exe'*
'Microsoft 365 Copilot Installer.exe'*
'Phone Link'/
'Pyhton Module 6.pptx'
'Python Module 5.pptx'
'Rays Of Hope.jpg'
'Telegram Desktop'/
UploadFromMobile/
VSCodeUserSetup-x64-1.109.0.exe*
'WhatsApp Image 2026-02-04 at 5.32.09 PM.jpeg'
'WhatsApp Image 2026-02-05 at 1.46.03 AM (1).jpeg'
'WhatsApp Image 2026-02-05 at 1.46.03 AM.jpeg'
'WhatsApp Installer.exe'*
desktop.ini
'download (1).jpg'
'download (2).jpg'
'download (3).jpg'
'download (4).jpg'
download.jpg
file.pdf
fujji.jpg
mingw-get-setup.exe*
node-v24.13.0-x64.msi
python-manager-25.2.msix
windhawk_setup.exe*

vijay@jarvis MINGW64 ~/downloads

```

File & Folder Commands

Command: touch file1.txt

Description: Creates a new empty file.

Screenshot:

```
MINGW64:/c/Users/vijay
vijay@jarvis MINGW64 ~
$ touch v1

vijay@jarvis MINGW64 ~
$ rm v1

vijay@jarvis MINGW64 ~
$ ls
AppData/
'Application Data'@
Contacts/
Cookies@
CrossDevice/
Documents/
Downloads/
Favorites/
Links/
'Local Settings'@
Music/
'My Documents'@
NTUSER.DAT
NTUSER.DAT{6e6239c7-0aea-11f1-85b4-c0a8109e56c3}.TM.blf
NTUSER.DAT{6e6239c7-0aea-11f1-85b4-c0a8109e56c3}.TMContainer00000000
1.regtrans-ms
NTUSER.DAT{6e6239c7-0aea-11f1-85b4-c0a8109e56c3}.TMContainer00000000
2.regtrans-ms
NetHood@
OneDrive/
PrintHood@
Recent@
'Saved Games' /
Searches/
SendTo@
'Start Menu'@
Templates@
Videos/
ntuser.dat.LOG1
ntuser.dat.LOG2
ntuser.ini

vijay@jarvis MINGW64 ~
$
```

Command: mkdir folder_name

Description: Creates a new folder.

Command: rm file1.txt

Description: Deletes a file.

Screenshot:

```
MINGW64:/c/Users/vijay
2. regtrans-ms
NetHood@
OneDrive/
PrintHood@
Recent@
'Saved Games'/
Searches/
SendTo@
'Start Menu'@
Templates@
Videos/
ntuser.dat.LOG1
ntuser.dat.LOG2
ntuser.ini

vijay@jarvis MINGW64 ~
$ mkdir vn1

vijay@jarvis MINGW64 ~
$ ls
AppData/
'Application Data'@
Contacts/
Cookies@
CrossDevice/
Documents/
Downloads/
Favorites/
Links/
'Local Settings'@
Music/
'My Documents'@
NTUSER.DAT
NTUSER.DAT{6e6239c7-0aea-11f1-85b4-c0a8109e56c3}.TM.b1f
NTUSER.DAT{6e6239c7-0aea-11f1-85b4-c0a8109e56c3}.TMContainer00000000000000000000
1. regtrans-ms
NTUSER.DAT{6e6239c7-0aea-11f1-85b4-c0a8109e56c3}.TMContainer00000000000000000000
2. regtrans-ms
NetHood@
OneDrive/
PrintHood@
Recent@
'Saved Games'/
Searches/
SendTo@
'Start Menu'@
Templates@
Videos/
ntuser.dat.LOG1
ntuser.dat.LOG2
ntuser.ini
vn1/

vijay@jarvis MINGW64 ~
```

File Content Commands

Command: cat file1.txt

Description: Displays file content in terminal.

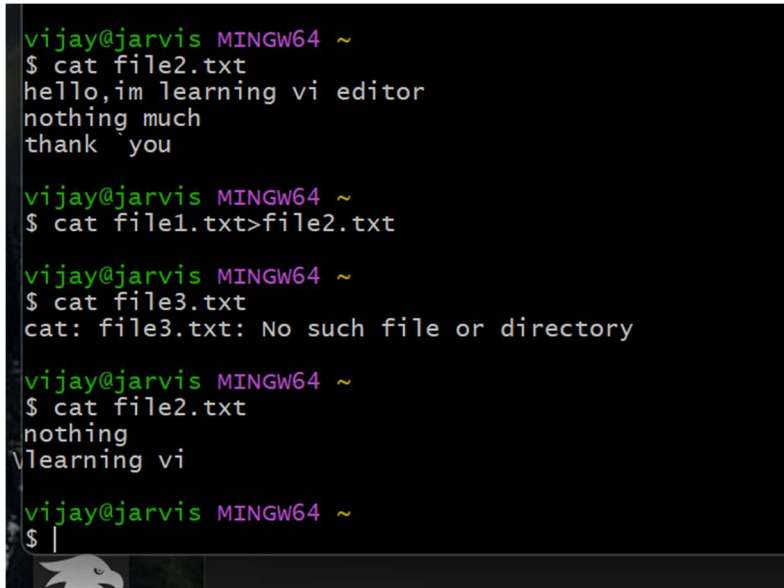
Command: echo "text" > file1.txt

Description: Writes text into a file.

Command: `echo "text" >> file1.txt`

Description: Appends text to file.

Command: `cat > file2.txt`

A terminal window with a black background and green text. The prompt is 'vijay@jarvis MINGW64 ~'. The user enters '\$ cat file2.txt' and the output is 'hello,im learning vi editor', 'nothing much', and 'thank you'. Then the user enters '\$ cat file1.txt>file2.txt'. Next, the user enters '\$ cat file3.txt' and receives the error 'cat: file3.txt: No such file or directory'. Finally, the user enters '\$ cat file2.txt' and the output is 'nothing' and 'learning vi'.

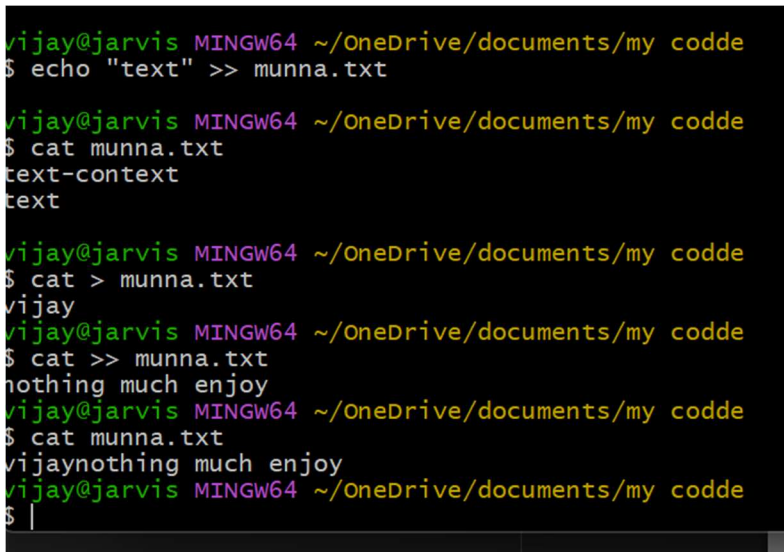
```
vijay@jarvis MINGW64 ~  
$ cat file2.txt  
hello,im learning vi editor  
nothing much  
thank you  
  
vijay@jarvis MINGW64 ~  
$ cat file1.txt>file2.txt  
  
vijay@jarvis MINGW64 ~  
$ cat file3.txt  
cat: file3.txt: No such file or directory  
  
vijay@jarvis MINGW64 ~  
$ cat file2.txt  
nothing  
learning vi  
  
vijay@jarvis MINGW64 ~  
$ |
```

Description: Creates file and allows manual input.

Command: `cat file1.txt > file2.txt`

Description: Copies content from one file to another.

Screenshot:

A terminal window with a black background and green text. The prompt is 'vijay@jarvis MINGW64 ~/OneDrive/documents/my codde'. The user enters '\$ echo "text" >> munna.txt'. Then the user enters '\$ cat munna.txt' and the output is 'text-context' and 'text'. Next, the user enters '\$ cat > munna.txt' and the prompt changes to 'vijay'. Then the user enters '\$ cat >> munna.txt' and the output is 'nothing much enjoy'. Finally, the user enters '\$ cat munna.txt' and the output is 'vijaynothing much enjoy'.

```
vijay@jarvis MINGW64 ~/OneDrive/documents/my codde  
$ echo "text" >> munna.txt  
  
vijay@jarvis MINGW64 ~/OneDrive/documents/my codde  
$ cat munna.txt  
text-context  
text  
  
vijay@jarvis MINGW64 ~/OneDrive/documents/my codde  
$ cat > munna.txt  
vijay  
vijay@jarvis MINGW64 ~/OneDrive/documents/my codde  
$ cat >> munna.txt  
nothing much enjoy  
vijay@jarvis MINGW64 ~/OneDrive/documents/my codde  
$ cat munna.txt  
vijaynothing much enjoy  
vijay@jarvis MINGW64 ~/OneDrive/documents/my codde  
$ |
```



```
vi@jay@jarvis MINGW64 ~/OneDrive/documents/my codde
$ cat munna.txt
1.Which planet is known as the "Red Planet"?
a = Venus
b = Mars
c = Jupiter
d = Saturn
Answer: b

vi@jay@jarvis MINGW64 ~/OneDrive/documents/my codde
$ echo "text-context" > munna.txt

vi@jay@jarvis MINGW64 ~/OneDrive/documents/my codde
$ ls
demo.c demo.cpp demo.exe* ex.csv exe.csv exe.file main.exe* munna.txt nothing.py

vi@jay@jarvis MINGW64 ~/OneDrive/documents/my codde
$ echo "text" >> munna.txt

vi@jay@jarvis MINGW64 ~/OneDrive/documents/my codde
$ cat munna.txt
text-context
text

vi@jay@jarvis MINGW64 ~/OneDrive/documents/my codde
$
```

Git + vi Editor Commands

Command: `git config --global core.editor "vi"`

Description: Sets vi as default Git editor.

Command: `git var GIT_EDITOR`

Description: Shows default editor used by Git.

Command: `vi file.txt`

Command: :wq

Description: Saves and exits vi editor.

Screenshot:

[illegible]

vi Editor Operations

Command: i / a / o / 0

Description: Insert text in different positions.

Screenshot:

Command: dd

Description: Delete a line.

Screenshot:

```
MINGW64:/c/Users/vijay  
new line  
vijay  
cse(btech)  
im a good boy  
cool guy who wants to chill everytime  
~  
~  
~  
~  
~  
~  
~  
~  
~  
~  
~  
~
```

Command: d\$

Description: Delete till end of line

Command: yy

Description: Copy a line.

Command: p / P

Description: Paste below or above.

Screenshot:

